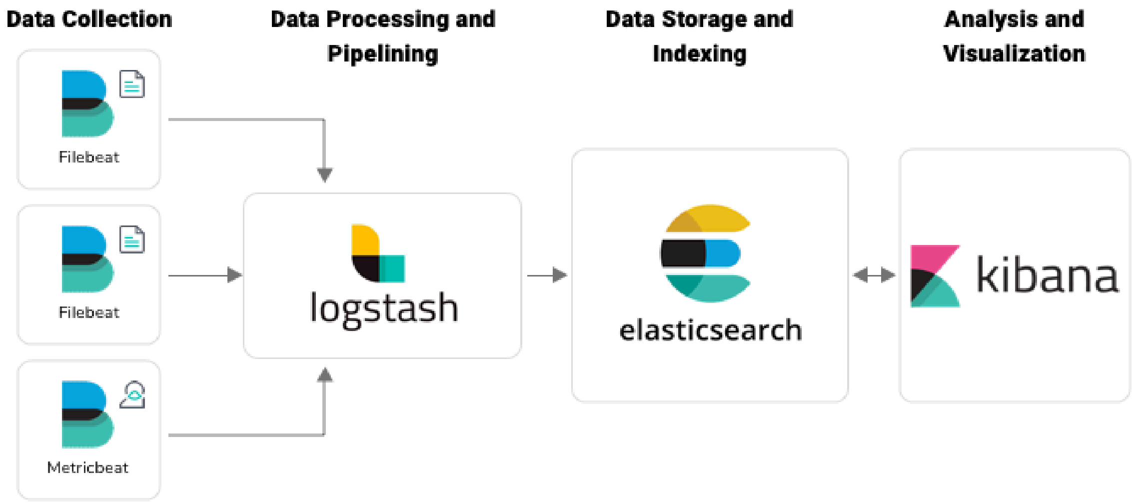
# Installation Elasticsearch logstash Kibana Nginx-filebeat

The Elastic Stack — formerly known as the ELK Stack — is a collection of open-source software produced by Elastic which allows you to search, analyze, and visualize logs generated from any source in any format, a practice known as centralized logging. Centralized logging can be useful when attempting to identify problems with your servers or applications as it allows you to search through all of your logs in a single place. It's also useful because it allows you to identify issues that span multiple servers by correlating their logs during a specific time frame.



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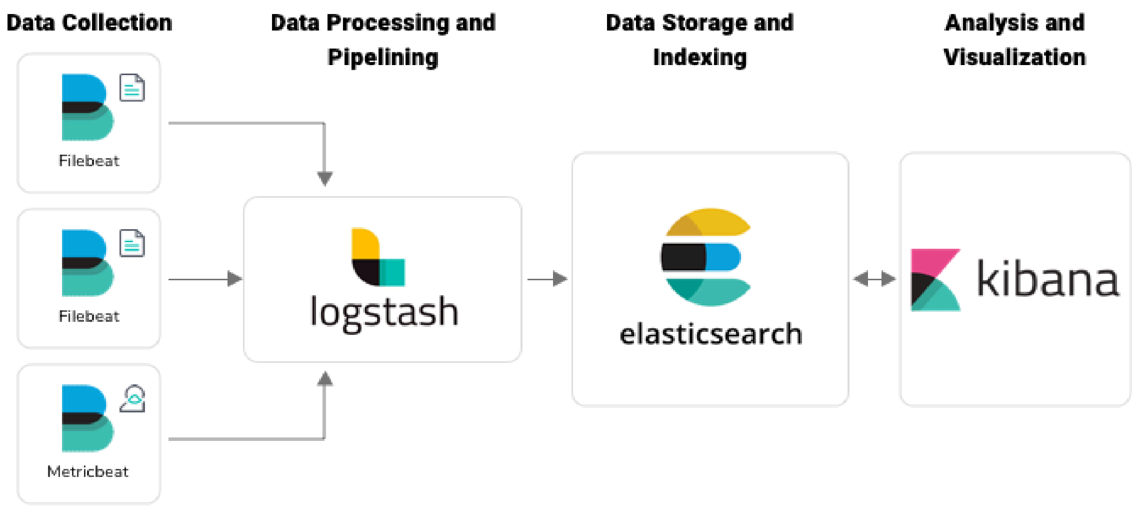
# Installation and Configure Elasticsearch

## STEP 1

### Elasticsearch Introduction

The Elastic Stack — formerly known as the ELK Stack — is a collection of open-source software produced by Elastic which allows you to search, analyze, and visualize logs generated from any source in any format, a practice known as centralized logging. Centralized logging can be useful when attempting to identify problems with your servers or applications as it allows you to search through all of your logs in a single place. It's also useful because it allows you to identify issues that span multiple servers by correlating their logs during a specific time frame.




Elasticsearch is the distributed search and analytics engine at the heart of the Elastic Stack. Logstash and Beats facilitate collecting, aggregating, and enriching your data and storing it in Elasticsearch. Kibana enables you to interactively explore, visualize, and share insights into your data and manage and monitor the stack. Elasticsearch is where the indexing, search, and analysis magic happens.

# STEP 2

## System requirements for Elasticsearch

| Component | Dev Environment Minimum | Production Minimum |
|---|---|---|
| CPU | 8 CPU Cores | 16 CPU cores |
| Memory | 16 GB RAM | 32 GB RAM |
| Storage | 250 GB SSD | 500 GB SSD |

# STEP 3

## Installation Java

Node : ELK Stack server

To install this version, first update the package index :

sudo apt update

Next, check if Java is already installed :

java -version

install the default Java Runtime Environment (JRE) :

sudo apt install default-jre

Verify the installation with :

java -version

You'll see output similar to the following:

Output

openjdk version "11.0.11" 2022-11-20

OpenJDK Runtime Environment (build 11.0.11+9-Ubuntu-0ubuntu2.20.04)

OpenJDK 64-Bit Server VM (build 11.0.11+9-Ubuntu-0ubuntu2.20.04, mixed mode, sharing))

# STEP 4

## Installation Elasticsearch

**Node : ELK Stack server**

Import the Elasticsearch PGP signing key :

curl -x "http://172.30.5.136:8118"  -fsSL https://artifacts.elastic.co/GPG-KEY-elasticsearch | sudo apt-key add -

Add Elasticsearch APT repository :

echo "deb https://artifacts.elastic.co/packages/7.x/apt stable main" | sudo tee -a /etc/apt/sources.list.d/elastic-7.x.list

Update the system :

sudo apt update

Install Elasticsearch :

sudo apt install elasticsearch

Start Elasticsearch service:

sudo systemctl start elasticsearch

Get status service :

sudo systemctl status elasticsearch

Enable Elasticsearch service to start at system startup:

sudo systemctl enable elasticsearch

# STEP 5

## Configure Elasticsearch

Node : ELK Stack server

The default path for the Elasticsearch configuration file is
**/etc/elasticsearch/elasticsearch.yml**

Edit Elasticsearch configuration file :

```
sudo nano /etc/elasticsearch/elasticsearch.yml
```

Elasticsearch listens for traffic from everywhere on port 9200

If you want it to listen only on a specific interface, you can specify its IP in place of localhost

network.host: "ELK server IP"

Uncomment the following lines :

```
network.host: "x.x.5.138"

http.port: "9200"

discovery.seed_hosts: ["x.x.5.138", "host2"]
```

Restart the Elasticsearch service :

```
sudo systemctl restart elasticsearch
```

Get status service :

```
sudo systemctl status elasticsearch
```

status = active (running)

Verify that Elasticsearch is running and listening on port 9200 :

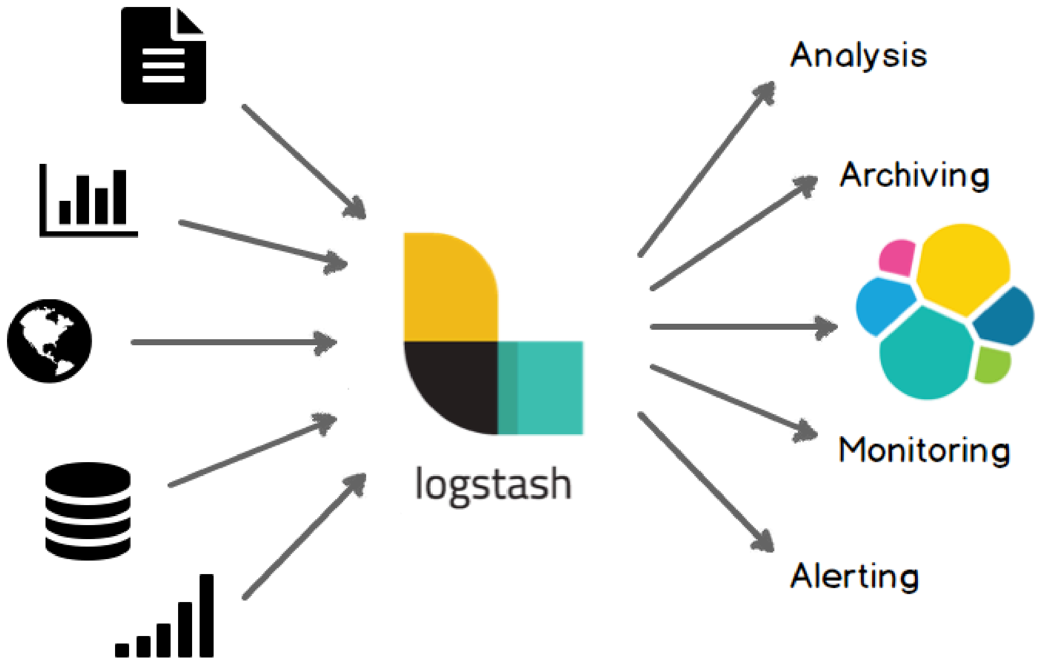
```
curl -X GET "x.x.5.138:9200"
```

# Installation and Configure Logstash

## STEP 6

### Logstash Introduction

Logstash is an open source data collection engine with real-time pipelining capabilities. Logstash can dynamically unify data from disparate sources and normalize the data into destinations of your choice. Cleanse and democratize all your data for diverse advanced downstream analytics and visualization use cases.




A Logstash pipeline has two required elements, input and output, and one optional element, filter. The input plugins consume data from a source, the filter plugins process the data, and the output plugins write the data to a destination.

# STEP 7

## Installation Logstash

**Node : ELK Stack server**

Update the system :

```
sudo apt update
```

Install logstash :

```
sudo apt install logstash
```

Start logstash service:

```
sudo systemctl start logstash
```

Verify Logstash service status :

```
sudo systemctl status logstash
```

Enable logstash service to start at system startup:

```
sudo systemctl enable logstash
```

# STEP 8

## Configure Logstash

### Node : ELK Stack server

Create a configuration file called 02-beats-input.conf where you will set up your Filebeat input:

```
sudo nano /etc/logstash/conf.d/02-beats-input.conf
```

Insert the following input configuration. This specifies a beats input that will listen on TCP port 5044.

```
input {
 beats {
  port => 5044
 }
}
```

Next, create a configuration file called 30-elasticsearch-output.conf :

```
sudo nano /etc/logstash/conf.d/30-elasticsearch-output.conf
```

Insert the following `output` configuration. Essentially, this output configures Logstash to store the Beats data in Elasticsearch, which is running at `172.30.5.138:9200`, in an index named after the Beat used. The Beat used in this project is Filebeat:

```
output {
 if [@metadata][pipeline] {
	elasticsearch {
	hosts => ["x.x.5.138:9200"]
	manage_template => false
	index => "%{[@metadata][beat]}-%{[@metadata][version]}-%{+YYYY.MM.dd}"
```

```
        pipeline => "%{[@metadata][pipeline]}"
        }
  } else {
        elasticsearch {
        hosts => ["x.x.5.138:9200"]
        manage_template => false
        index => "%{[@metadata][beat]}-%{[@metadata][version]}-%{+YYYY.MM.dd}"
        }
  }
}
```

## Test your Logstash configuration with this command :

sudo -u logstash /usr/share/logstash/bin/logstash --path.settings /etc/logstash -t

If there are no syntax errors, your output will display Config Validation Result: OK. Exiting Logstash after a few seconds

restart the Logstash service :

sudo systemctl restart logstash

## Verify Logstash service status :

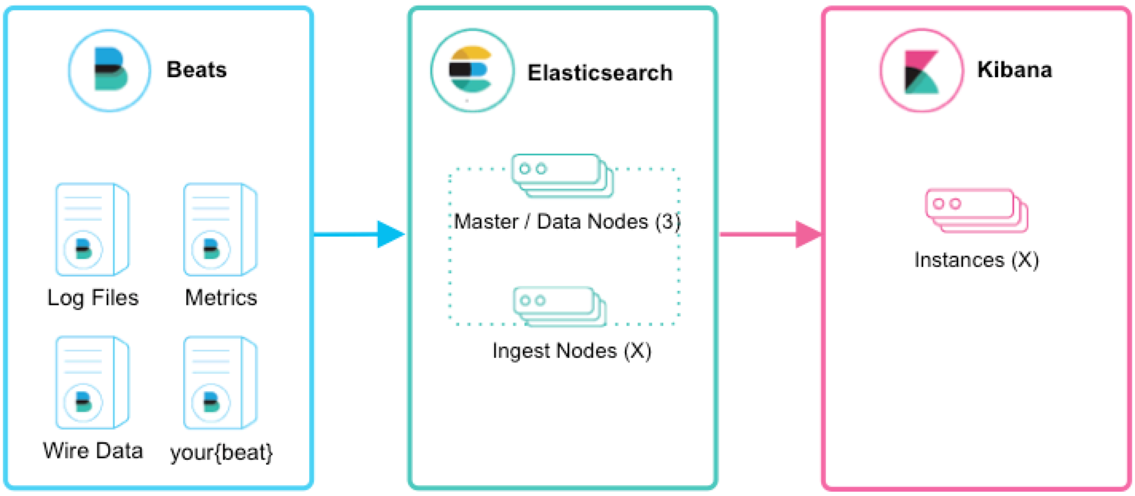
sudo systemctl status logstash

# Installation and Configure Kibana

# STEP 9

## Kibana Introduction

Kibana is an free and open frontend application that sits on top of the Elastic Stack, providing search and data visualization capabilities for data indexed in Elasticsearch. Commonly known as the charting tool for the Elastic Stack (previously referred to as the ELK Stack after Elasticsearch, Logstash, and Kibana), Kibana also acts as the user interface for monitoring, managing, and securing an Elastic Stack cluster — as well as the centralized hub for built-in solutions developed on the Elastic Stack.



# STEP 10

## Installation Kibana

**Node : ELK Stack server**

### Update the system :

```
sudo apt update
```

### Install kibana :

```
sudo apt install kibana
```

### Start kibana service:

```
sudo systemctl start kibana
```

### Verify kibana service status :

```
sudo systemctl status kibana
```

### Enable kibana service to start at system startup:

```
sudo systemctl enable kibana
```

# STEP 11

## Configure Kibana

**Node : ELK Stack server**

The default path for the kibana configuration file is **/etc/kibana/kibana.yml**

Edit kibana configuration file :

```
sudo nano /etc/kibana/kibana.yml
```

Uncomment the following lines :

```
server.port: 5601
server.host: "localhost"
server.name: " ansibleserver-Virtual-Machine"
elasticsearch.hosts: ["http://x.x.5.138:9200"]
```

Restart the kibana service :

```
sudo systemctl restart kibana
```

Get status service :

```
sudo systemctl status kibana
```

status = active (running)

# Installation Nginx and Configure for kibana

# STEP 12

## Installation Nginx

**Node : ELK Stack server**

Because Kibana is configured to only listen on localhost, we must set up a reverse proxy to allow external access to it. We will use Nginx for this purpose, which should already be installed on your server.

Update the system :

```
sudo apt update
```

Install nginx :

```
sudo apt install nginx
```

Verify nginx service status :

```
sudo systemctl status nginx
```

Enable nginx service to start at system startup:

```
sudo systemctl enable nginx
```

# STEP 13

## Configure Nginx for Kibana

Node : ELK Stack server

Using nano or your preferred text editor, create the Nginx server block file:

sudo nano /etc/nginx/sites-available/kibana-test

```
server {
  listen 80;
  server_name kibana-test.local;
  location / {
    proxy_pass http://localhost:5601;
    proxy_http_version 1.1;
    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection 'upgrade';
    proxy_set_header Host $host;
    proxy_cache_bypass $http_upgrade;
  }
}
```

enable the new configuration by creating a symbolic link to the sites-enabled directory :

sudo ln -s /etc/nginx/sites-available/kibana-test /etc/nginx/sites-enabled/kibana-test

Then check the configuration for syntax errors:

sudo nginx -t

restart the Nginx service:

sudo systemctl reload nginx

# STEP 14

## Visit the Kibana dashboard

Setting host file :

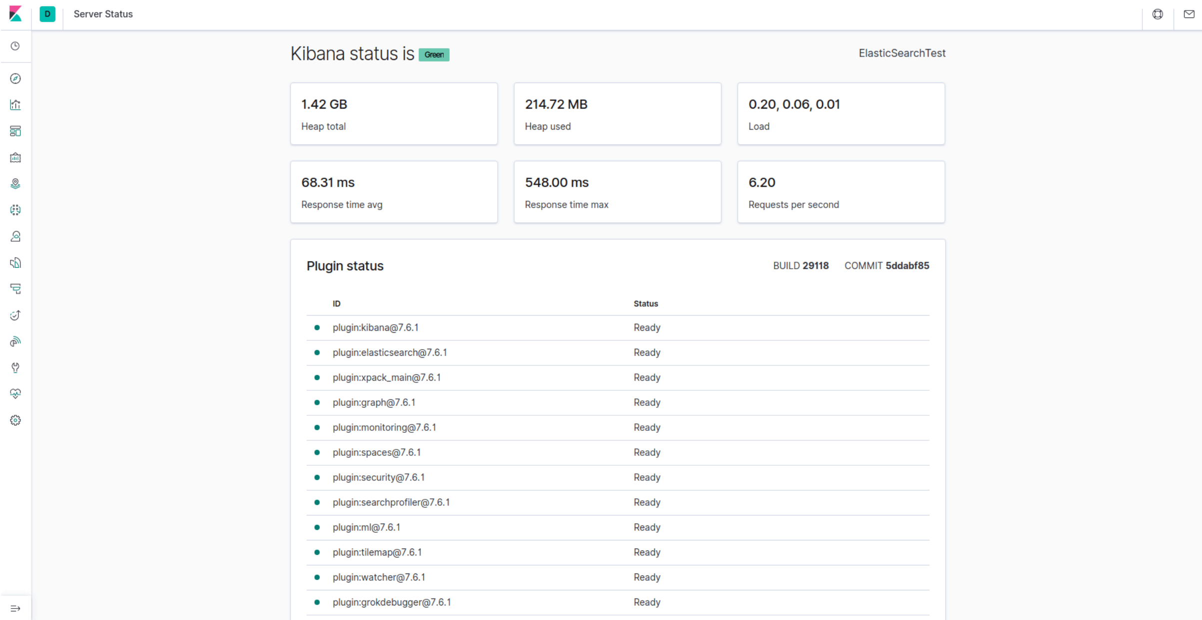
sudo nano /etc/hosts

kibana-morsa.local        ELK Stack server IP

kibana-morsa.local        x.x.5.138

You can check the Kibana server's status page by navigating to the following address :

http://kibana-test.local/status




This status page displays information about the server's resource usage and lists the installed plugins.

# Installation Nginx and Configure for log management

# STEP 15

### Installation Nginx on remote server

***Node : a-kube-master***

Update the system :

```
sudo apt update
```

Install nginx :

```
sudo apt install nginx
```

Verify nginx service status :

```
sudo systemctl status nginx
```

Enable nginx service to start at system startup:

```
sudo systemctl enable nginx
```

# STEP 16

## Configure Nginx

Node : a-kube-master

Create the directory for your_domain :

sudo mkdir -p /var/www/test-x/html

assign ownership of the directory with the $USER environment variable :

sudo chown -R $USER:$USER /var/www/test-x/html

sudo chmod -R 755 /var/www/test-x

create a sample index.html page :

sudo nano /var/www/test-x/html/index.html

```
<html>
    <head>
        <title>Welcome to test-x!</title>
    </head>
    <body>
        <h1>Success!  The test-x server block is working!</h1>
    </body>
</html>
```

In order for Nginx to serve this content, it’s necessary to create a server block with the correct directives.

sudo nano /etc/nginx/sites-available/test-x

Paste in the following configuration block :

```
server {
    listen 80;
    listen [::]:80;
```

```
        root /var/www/test-x/html;
        index index.html index.htm index.nginx-debian.html;

        server_name test-x.local www.test-x.local;

        location / {
                try_files $uri $uri/ =404;
        }
}
```

enable the file by creating a link from it to the sites-enabled directory :

sudo ln -s /etc/nginx/sites-available/test-x /etc/nginx/sites-enabled/

Test to make sure that there are no syntax errors in any of your Nginx files:

sudo nginx -t

If there aren't any problems, restart Nginx to enable your changes:

sudo systemctl restart nginx

setting host file :

sudo nano /etc/hosts

test-x.local        a-kube-master IP

Nginx should now be serving your domain name. You can test this by navigating to http://test-x.local , where you should see something like this:

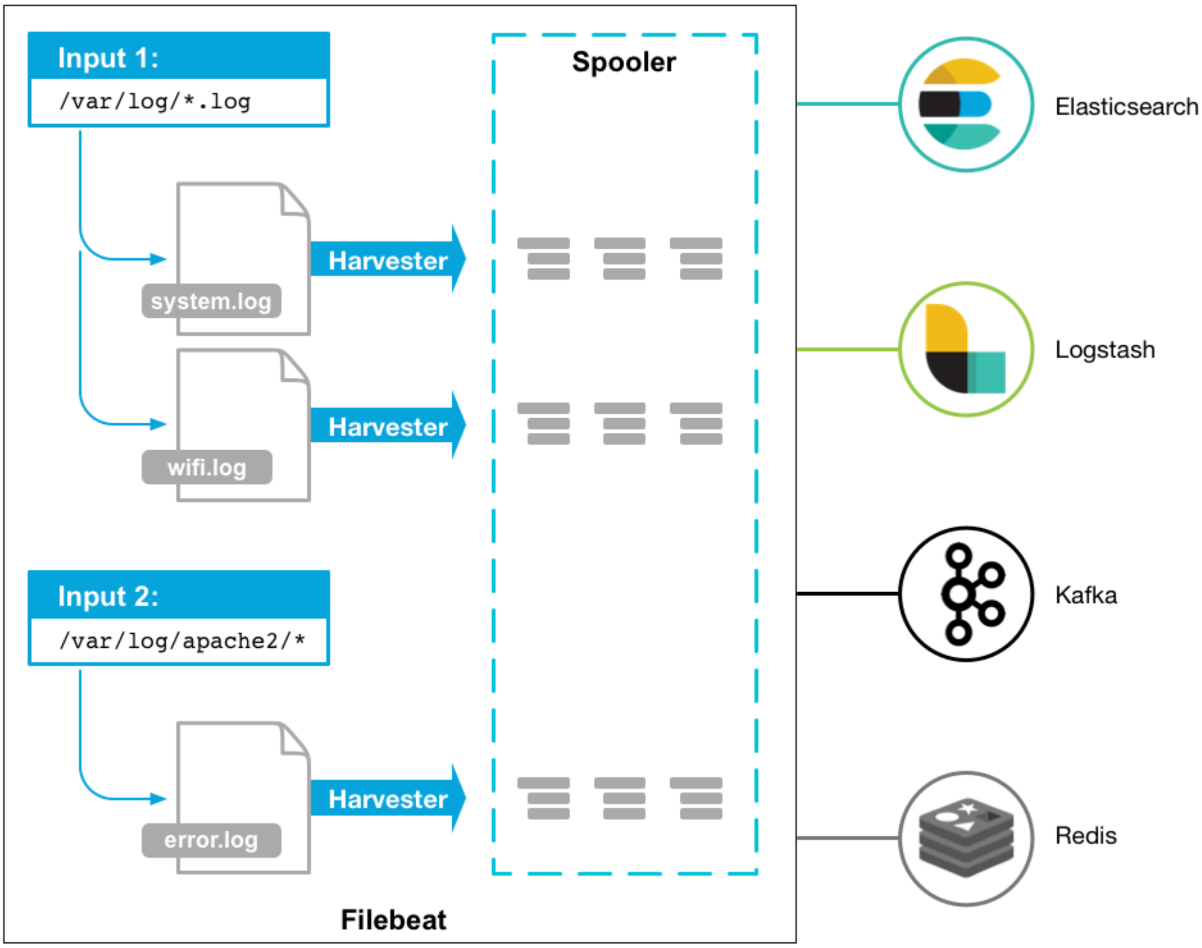
```
Success!  The test-x server block is working!
```

## Installation and Configure Filebeat

# STEP 17

### Filebeat Introduction

Filebeat is a lightweight shipper for forwarding and centralizing log data. Installed as an agent on your servers, Filebeat monitors the log files or locations that you specify, collects log events, and forwards them either to Elasticsearch or Logstash for indexing.



# STEP 18

## Installation Filebeat

### Node : a-kube-master

Install Curl & ca-certificates:

sudo apt install curl ca-certificates

Import the filebeat PGP signing key :

curl -x "http://172.30.5.136:8118"  -fsSL https://artifacts.elastic.co/GPG-KEY-elasticsearch | sudo apt-key add -

Add filebeat APT repository :

echo "deb https://artifacts.elastic.co/packages/7.x/apt stable main" | sudo tee -a /etc/apt/sources.list.d/elastic-7.x.list

Update the system :

sudo apt update

Install filebeat :

sudo apt install filebeat

Start filebeat service:

sudo systemctl start filebeat

Get status service :

sudo systemctl status filebeat

Enable filebeat service to start at system startup:

sudo systemctl enable filebeat

# STEP 19

## Configure Filebeat

Node : a-kube-master

The default path for the filebeat configuration file is **/etc/filebeat/filebeat.yml**

Edit filebeat configuration file :

sudo nano /etc/filebeat/filebeat.yml

Uncomment the following lines :

output.elasticsearch:

hosts: ["x.x.5.138:9200"]

username: "ansible-server"

password: "x"

Restart the filebeat service :

sudo systemctl restart filebeat

Get status service :

sudo systemctl status filebeat

status = active (running)

# STEP 20

## Enable Nginx module

Node : a-kube-master

Enable the Filebeat nginx module :

sudo filebeat modules enable nginx
You can see a list of enabled and disabled modules by running :

sudo filebeat modules list

we need to set up the Filebeat ingest pipelines, which parse the log data before sending it through logstash to Elasticsearch

sudo filebeat setup --pipelines --modules nginx

load the index template into Elasticsearch :

sudo filebeat setup --index-management -E output.logstash.enabled=false -E 'output.elasticsearch.hosts=["x.x.5.138:9200"]'

Output
Index setup finished.

**config nginx module:**

sudo nano /etc/filebeat/modules.d/nginx.yml

var.paths: ["/var/log/nginx/*.log"]

Now you can restart and enable Filebeat:

sudo systemctl restart filebeat

sudo systemctl enable filebeat

Verify that Filebeat is shipping log files to Logstash for processing :

curl -XGET 'http://x.x.5.138:9200/filebeat-*/_search?pretty'

# Create Dashboard and Visualization

# STEP 21

## Create index pattern

Open your web browser and access the Kibana web interface using the URL

http://kibana-test.local

create  index pattern :

Click the "Stack Management → index patterns → create index pattern "

Name : filebeat-*

Timestamp field : Timestamp

# STEP 22

## Create dashboard for nginx logs

Create Dashboard nginx :

Click the  Dashboard → create visualization

Index pattern : filebeat-*

Available fields :

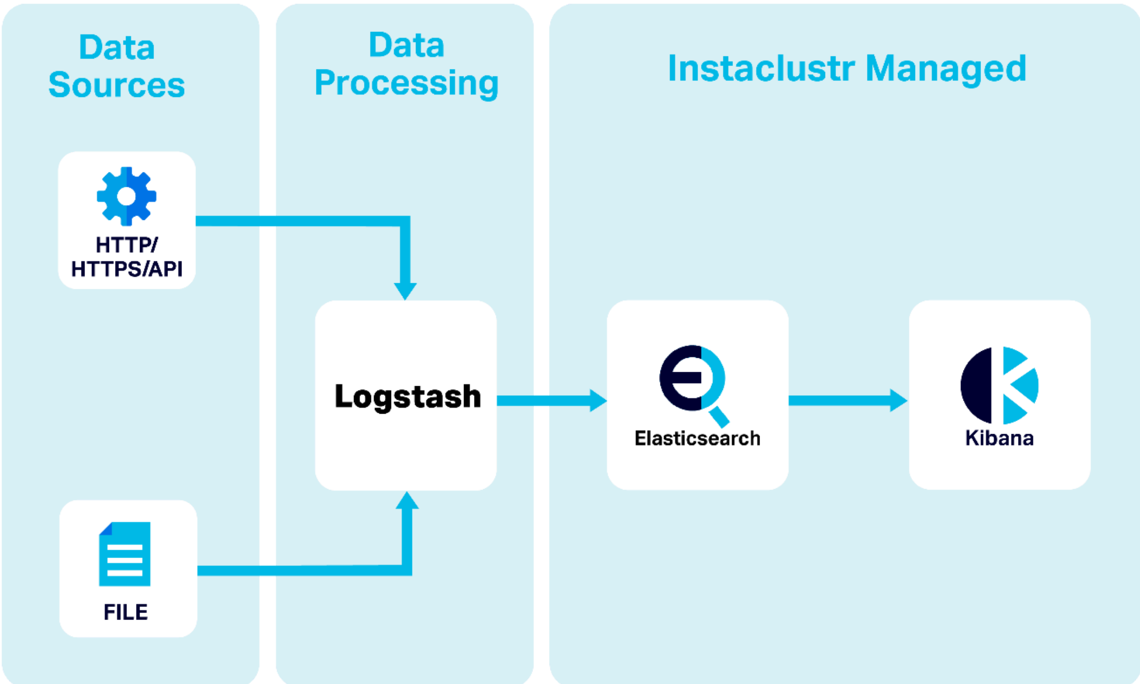
1. Timestamp
2. http.request.referrer
3. url.original
4. Count of records
5. http.request.method
6. host.os.version
7. fileset.name
8. host.hostname
9. host.os.family
10. log.file.path

## Receive Logs through HTTP requests

# STEP 23

### HTTP-Input-Plugin Introduction

Using this input you can receive single or multiline events over http(s). Applications can send an HTTP request to the endpoint started by this input and Logstash will convert it into an event for subsequent processing. Users can pass plain text, JSON, or any formatted data and use a corresponding codec with this input. For Content-Type application/json the json codec is used, but for all other data formats, plain codec is used.



# STEP 24

## Installation Plugin logstash-input-http

Node : ELK stack

To list the plugins currently available in your deployment:

cd /usr/share/logstash

sudo bin/logstash-plugin list

### Install Plugin logstash-input-http

sudo bin/logstash-plugin install logstash-input-http

After a plugin is successfully installed, you can use it in your configuration file.

### Update Plugin logstash-input-http

Plugins have their own release cycles and are often released independently of Logstash's core release cycle. Using the update subcommand you can get the latest version of the plugin.

sudo bin/logstash-plugin update  logstash-input-http

# STEP 25

## Configure Logstash for Http requests

Node : ELK stack

When you configure this plugin in the input section, it will launch a HTTP server and create events from requests sent to this endpoint

cd /etc/logstash/conf.d

**Delete previous configuration files :**

sudo rm 02-beats-input.conf

sudo rm 30-elasticsearch-output.conf

**create a configuration file logstash.conf:**

By default it will bind the web server to all hosts ("0.0.0.0") and open the TCP port 8080

Input config : Logstash (0.0.0.0:8080)

Output config : Elasticsearch (x.x.5.138:9200)

sudo nano logstash.conf

```
input {
 http {
  host => "0.0.0.0" # default: 0.0.0.0
  port => "8080" # default: 8080
 }
}

output {
#  stdout {
#    codec => rubydebug}

 elasticsearch {
  hosts => ["172.30.5.138:9200"]
  manage_template => false
  index => "logstash2"

 }
}
```

Test your Logstash configuration with this command :

sudo -u logstash /usr/share/logstash/bin/logstash --path.settings /etc/logstash -t

If there are no syntax errors, your output will display Config Validation Result: OK. Exiting Logstash after a few seconds

restart the Logstash service :

sudo systemctl restart logstash

Verify Logstash service status :

sudo systemctl status logstash

# STEP 26

## Send Logs to logstash using HTTP Request

Node : ELK stack

Send Logs using Curl :

```
curl -XPUT 'http://x.x.5.138:8080/twitter/tweet/1' -d 'hello'
```

```
curl -H "content-type: application/json" -XPUT 'http://x.x.5.138:8080/twitter/tweet/1' -d '{
    "user" : "kimchy",
    "post_date" : "2009-11-15T14:12:12",
    "message" : "trying out Elasticsearch"
}'
```

# STEP 27

## Create index pattern

Open your web browser and access the Kibana web interface using the URL

http://kibana-test.local

create  index pattern :

Click the “Stack Management ⍰ index patterns ⍰ create index pattern ”

Name : logstash2

Timestamp field : Timestamp

# STEP 28

## Create dashboard for nginx logs

Create Dashboard nginx :

Click the  Dashboard ⍰ create visualization

Index pattern : logstash2

Available fields :

1. Timestamp
2. headers.content_type.keyword
3. headers.http_host.keyword
4. message.keyword
5. user.keyword
6. host.keyword

## Successful

## END